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Grade	**100.00** out of 100.00
Name	DHIRU BHAI 730423104017

Question 1

Correct

Mark 10.00 out of 10.00

Write a program to display hte given pattern

For example:

Input	Result

3	`  *`
	`* *`
	`* * *`

4	`   *`
	`  * *`
	`* * *`
	`* * * *`

Answer: (penalty regime: 0 %)

Language java

```
import java.util.Scanner;
public class CenteredAsteriskPattern {
    public static void main(String [] args)
    {
        Scanner sc =new Scanner(System.in);
        int rows =sc.nextInt();
        int k=2*rows-2;
        for(int i=0;i<rows;i++){
            for(int j=0; j<k; j++){
              System.out.print(" ");
            }
            k=k-2;
            for(int j=0;j<i+1;j++){
                System.out.print("* ");
            }
            System.out.println();
        }
        System.out.println();
    }
}
```

	Input	Expected	Got	

✔	3	`  *`
		`* *`
		`* * *`
			`  *`
			`* *`
			`* * *`
				✔

✔	4	`   *`
		`  * *`
		`* * *`
		`* * * *`
			`   *`
			`  * *`
			`* * *`
			`* * * *`
				✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 2

Correct

Mark 10.00 out of 10.00

Write a program to print the following pattern for a given value of N

Input
3
a
Output
a
aaa
aaaaa
aaa
a
Input
6
X
Output

```
     X
    XXX
   XXXXX
  XXXXXXX
 XXXXXXXXX
XXXXXXXXXXX
 XXXXXXXXX
  XXXXXXX
   XXXXX
    XXX
     X
```

For example:

Input	Result
3 a	a aaa aaaaa aaa a

Input	Result
6 X	X XXX XXXXX XXXXXXX XXXXXXXXX XXXXXXXXXXX XXXXXXXXX XXXXXXX XXXXX XXX X

Answer: (penalty regime: 0 %)

Language java

```
import java.util.Scanner;
public class Pattern
{
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);
        int  n= sc.nextInt();
        char ch=sc.next().charAt(0);
        printPattern(n,ch);
    }
    public static void printPattern(int n, char ch)
    {
        int spaces=n-1;
        int stars=1;
        for(int i=1;i<=n;i++)
        {
            for(int j=1;j<=spaces;j++)
            {
                System.out.print(" ");
            }
                for(int j=1;j<=stars;j++)
                {
                    System.out.print(ch);
                }
                System.out.println();
                spaces--;
                stars+=2;
            }
            spaces=1;
            stars=n*2-3;
            for(int i=1;i<=n-1;i++)
            {
                for(int j=1;j<=spaces;j++)
                {
                    System.out.print(" ");
                }
                for(int j=1;j<=stars;j++)
                {
                    System.out.print(ch);
                }
                System.out.println();
                spaces++;
                stars-=2;
            }
        }
}
```

	Input	Expected	Got	
✔	3 a	a aaa aaaaa aaa a	a aaa aaaaa aaa a	✔
✔	6 X	X XXX XXXXX XXXXXXX XXXXXXXXX XXXXXXXXXXX XXXXXXXXX XXXXXXX XXXXX XXX X	X XXX XXXXX XXXXXXX XXXXXXXXX XXXXXXXXXXX XXXXXXXXX XXXXXXX XXXXX XXX X	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 3

Correct

Mark 10.00 out of 10.00

Write a program to rotate an array in anti-clockwise direction for the given number of rotation.

For example:

Input	Result
5 1 2 3 4 5 2	4 5 1 2 3
6 4 6 7 2 3 9 3	2 3 9 4 6 7
1 1 1	Invalid Input

Answer: (penalty regime: 0 %)

```
import java.util.Arrays;
import java.util.Scanner;
public class Main
{
    public static void main(String[] args)
    {
        Scanner s=new Scanner(System.in);
        int g=s.nextInt();
        int[] array=new int[g];
        for(int i=0;i<g;i++)
        {
            array[i]=s.nextInt();
        }
        int r=s.nextInt();
        if(g==1)
        {
            System.out.println("Invalid Input");
        }
        else
        {
            rightRotate(array,r);
            printArray(array);
        }
    }
    public static void rightRotate(int[]array,int k)
    {
        int n=array.length;
        k%=n;
        reverse(array,0,n-1);
        reverse(array,0,k-1);
        reverse(array,k,n-1);
    }
    public static void reverse(int [] array, int start,int end)
    {
        while(start<end)
        {
            int t=array[start];
            array[start]=array[end];
            array[end]=t;
            start++;
```

```
            end--;
        }
    }
    public static void printArray(int[] array)
    {
        for(int i=0;i<array.length;i++)
        {
            if(i>0)
            {
                System.out.print(" ");
            }
            System.out.print(array[i]);
            }
            System.out.println();
        }
}
```

	Input	Expected	Got	
✔	5 1 2 3 4 5 2	4 5 1 2 3	4 5 1 2 3	✔
✔	6 4 6 7 2 3 9 3	2 3 9 4 6 7	2 3 9 4 6 7	✔
✔	1 1 1	Invalid Input	Invalid Input	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 4

Correct

Mark 10.00 out of 10.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. If no majority element occurs in the array return 0

Input

9 <- number of elements

1
2
3
4
4
4
4
3
4

Output

4

For example:

Input	Result
9 1 2 3 4 4 4 4 3 4	4
9 1 2 3 3 3 4 4 5 4	0

Answer: (penalty regime: 0 %)

Language java

```
import java.util.*;
public class MajorityElement
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
       int[] nums=new int[n];
       for(int i=0;i<n;i++)
       {
           nums[i]=sc.nextInt();
       }
```

```
        System.out.println(findMajorityElement(nums));
    }
public static int findMajorityElement(int[] nums)
{
    Map<Integer, Integer>countMap=new HashMap<>();
    int majorityCount=nums.length/2;
    for(int num:nums)
    {
       countMap.put(num,countMap.getOrDefault(num,0)+1);
       if(countMap.get(num)>majorityCount)
       {
           return num;
       }
    }
    return 0;
}
    }

```

	Input	Expected	Got	
✔	9 1 2 3 4 4 4 4 3 4	4	4	✔
✔	9 1 2 3 3 3 4 4 5 4	0	0	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 5

Correct

Mark 10.00 out of 10.00

Stock Price[] is an array containing stock price of Y company for n days starting from day 1 till nth data. An investor can only buy and sell once. Write a function which will print day of buying and day of selling so that investor could have maximized his profit.

Sample Input 1

```
4
250
320
190
300
```

Sample Output 1

```
Day of buying : 3
Day of Selling : 4
```

For example:

Input	Result
4 250 320 190 300	Day of buying : 3 Day of Selling : 4
5 150 100 230 400 240	Day of buying : 2 Day of Selling : 4

Answer: (penalty regime: 0 %)

```
import java.util.*;
public class StockProfit
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[]prices=new int[n];
        for(int i=0;i<n;i++)
        {
            prices[i]=sc.nextInt();
        }
        maximizeProfit(prices);
        }
        public static void maximizeProfit(int[] prices)
        {
            int n=prices.length;
            if(n<2) return;

            int minPriceIndex=0;
            int maxProfit=0;
            int buyDay=0,sellDay=0;
            for(int i=1;i<n;i++)
```

```
            {
                if(prices[i]<prices[minPriceIndex])
                {
                    minPriceIndex=i;
                }
                int profit = prices[i]-prices[minPriceIndex];
                if(profit>maxProfit)
                {
                    maxProfit=profit;
                    buyDay=minPriceIndex+1;
                    sellDay=i+1;
                }
            }
            System.out.println("Day of buying : "+buyDay);
            System.out.println("Day of Selling : "+sellDay);
        }
    }
```

	Input	Expected	Got	
✔	4 250 320 190 300	Day of buying : 3 Day of Selling : 4	Day of buying : 3 Day of Selling : 4	✔
✔	5 150 100 230 400 240	Day of buying : 2 Day of Selling : 4	Day of buying : 2 Day of Selling : 4	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 6

Correct

Mark 10.00 out of 10.00

Given three arrays sorted in non-decreasing order, Print all common elements in these arrays.
Note: If the array has "no common elements", then print "NO Elements"

Constraints

1<=N1,N2,N3<=100

Sample Input 0

```
6
1 5 10 20 40 80
5
6 7 20 80 100
8
3 4 15 20 30 70 80 120
```

Sample Output 0

```
20 80
```

Answer: (penalty regime: 0 %)

```
import java.util.*;
public class CommonElements
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int n1=sc.nextInt();
        int[] arr1=new int[n1];
        for(int i=0;i<n1;i++)
        {
            arr1[i]=sc.nextInt();
        }
        int n2=sc.nextInt();
        int[] arr2=new int[n2];
        for(int i=0;i<n2;i++)
        {
            arr2[i]=sc.nextInt();
        }
        int n3=sc.nextInt();
        int []arr3 =new int[n3];
        for(int i=0;i<n3;i++)
        {
            arr3[i]=sc.nextInt();
        }
        int i=0,j=0,k=0;
        boolean found =false;
        while(i<n1 && j<n2 && k<n3)
        {
            if(arr1[i]==arr2[j] && arr2[j]==arr3[k])
            {
                System.out.print(arr1[i] + " ");
                found =true;
                i++;
                j++;
                k++;

            }
            else if(arr1[i]<arr2[j])
            {
```

```
                i++;
            }
            else if(arr2[j]<arr3[k])
            {
                j++;
            }
            else
            {
                k++;
            }
        }
        if(!found)
        {
            System.out.println("NO Elements");
        }

    }
}
```

	Input	Expected	Got	
✔	6 1 5 10 20 40 80 5 6 7 20 80 100 8 3 4 15 20 30 70 80 120	20 80	20 80	✔
✔	3 1 2 3 4 4 5 6 7 5 8 9 10 11 12	NO Elements	NO Elements	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 7

Correct

Mark 10.00 out of 10.00

Write a program to print Strong numbers between 1 to n.

Strong number is a special number whose sum of factorial of digits is equal to the original number. For example: 145 is strong number. Since, `1! + 4! + 5! = 145`

Example

Input

```
Input upper limit: 1000
```

Output

```
Strong numbers between 1-1000:
1, 2, 145
```

Input 1

```
1000
```

Output 1

```
Strong numbers between 1-1000:
1 2 145
```

For example:

Input	Result
100	1 2
500	1 2 145

Answer: (penalty regime: 0 %)

Language java

```
import java.util.*;
public class StrongNumber
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        for(int i=1;i<=n;i++)
        {
            if(isStrong(i))
            {
                System.out.print(i + " ");
            }
        }
    }
    public static boolean isStrong(int number)
    {
        int sum=0;
        int temp=number;
        while(temp>0)
        {
            int digit=temp%10;
            sum+=factorial(digit);
            temp/=10;
        }
```

```
        return sum==number;
    }
    public static int factorial(int num)
    {
        if(num==0|| num==1)
        {
            return 1;
        }
        int fact=1;
        for(int i=2;i<=num;i++)
        {
            fact*=i;
        }
        return fact;
    }
    }
```

	Input	Expected	Got	
✔	100	1 2	1 2	✔
✔	500	1 2 145	1 2 145	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 8

Correct

Mark 10.00 out of 10.00

A string is considered as Happy String if each of its given Vowels occur even number of times.

For example:

Input	Result
abba	HAPPY
abbe	NOT HAPPY

Answer: (penalty regime: 0 %)

Language java

```
import java.util.*;
public class HappyStringChecker
{
    public static void main(String [] args)
    {
        Scanner sc=new Scanner (System.in);
        String input=sc.nextLine();
        System.out.println(isHappyString(input) ? "HAPPY" : "NOT HAPPY");
    }
    private static boolean isHappyString(String str)
    {
        Map<Character,Integer>vowelCount=new HashMap<>();
        for(char c : str.toLowerCase().toCharArray())
        {
            if("aeiou".indexOf(c)!=-1)
            {
                vowelCount.put(c,vowelCount.getOrDefault(c, 0)+1);
            }
        }
        for(int count : vowelCount.values())
        {
            if(count%2!=0)
            {
                return false;
            }
        }
        return true;
    }
    }

```

	Input	Expected	Got	
✔	abba	HAPPY	HAPPY	✔
✔	abbe	NOT HAPPY	NOT HAPPY	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 9

Correct

Mark 10.00 out of 10.00

Professor gave him a string s. He taught about pure string. Professor assigned an assignment to Hariharan. The assignment is Hariharan needs to determine the length of the highest pure string which is a substring of s.

Note: Pure String is a string that only consists of vowels [upper or lower case]

Input format

The input contains a string s.

Output format

The output prints an integer denoting the length of the highest pure substring, that is substring comprises only vowels.

For example:

Input	Result
abcaac	3
bcd	0

Answer: (penalty regime: 0 %)

Language java

```
import java.util.Scanner;
public class Main{
    static boolean isVowel(char ch){
        ch= Character.toUpperCase(ch);
        return (ch=='A'|| ch=='E'||ch=='I'||ch=='O'||ch=='U');
    }
    static int countVowels(String str){
        int count=0;
        for(int i=0;i<str.length();i++)
        if(isVowel(str.charAt(i)))
        ++count;
        return count;
    }
    public static void main(String[] args)
{
    Scanner sc=new Scanner(System.in);
    String str =sc.nextLine();
    System.out.println(countVowels(str));
}
        }
```

	Input	Expected	Got	
✔	abcaac	3	3	✔
✔	bcd	0	0	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 10

Correct

Mark 10.00 out of 10.00

Write a program to check whether a string is a Palindrome or not

Constraints

Length of the string should be greater than one and less than 100.

For example:

Input	Result
madam	Palindrome
extend	Not a palindrome

Answer: (penalty regime: 0 %)

Language java

```
import java.util.*;
public class PalindromeChecker
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        String input=sc.nextLine();
        String reversed=new StringBuilder(input).reverse().toString();
        if(input.equals(reversed))
        {
            System.out.println("Palindrome");
        }
        else
        {
            System.out.println("Not a palindrome");
        }
    }
}
```

	Input	Expected	Got	
✔	madam	Palindrome	Palindrome	✔
✔	extend	Not a palindrome	Not a palindrome	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.